

CAIR Test Flow

ClaudMD / Aeliusmd → CAIR2 HL7 VXU | Sithum Dev DB | Vendor Org SF-013259 | Updated Sep 2026

1. What We Done

- Built a Python CAIR integration that reads vaccination data from ClaudMD DB, builds HL7 VXU messages, and sends them to CAIR2 over SOAP (submitSingleMessage).
- Synced Sithum dev DB (ClaudMD_Development_Sithum) with QA_2 so all CAIR tables/columns match.
- Configured .env for Sithum dev using CLINIC_DB_* (direct clinic DB connection).
- Created scripts/seed_sithum_cair_test_data.py and scripts/retry_submission.py for testing.
- Seeded 7 test submissions in Sithum DB; live sends to CAIR onboarding endpoint completed.
- Fixed SOAP client to SOAP 1.2 per CAIR WSDL (resolved HTTP 500: SOAP11 binding disabled).
- Received SOAP login for SF-013259; credentials set in .env and login verified (HTTP 200).
- Updated HL7 builder from CAIR ACK errors: race/ethnicity (DataGroups), address sanitize, provider NPI, dashed NDC, PD1=N, OBX funding source, MSH-22/RXA-11.4 clinic site org.
- Updated Excel Modification column in Required Fields mapping workbook for implemented fields.
- Confirmed clinic site org for MSH-22 / RXA-11.4: SF-012218 (PROVIDER_ORG_ID). Vendor SF-013259 and sample CA0054321 were rejected; SF-012218 accepted (no org blocking error).
- Mapped provider professional suffix (ORC-12.21 / RXA-10.21): prefer Providers.Degree; if Degree empty, use Title only when it is a short credential (e.g. MD).
- Latest ACK for submission **3** is **MSA|AA** (success) — only informational RXA-6 amount note remains.

2. How Background Job Runs

- Run **python cair_service.py** — one long-running process for all clinics.
- Every 5 minutes it wakes up, checks for pending work, processes a batch, then sleeps again.
- In dev (Sithum): connects directly to one clinic DB from CLINIC_DB_*.
- In QA/prod: reads ClinicSetup from master DB, then connects to each active clinic DB.
- For each pending row: read DB → build HL7 → send SOAP → update status in the same table.
- Up to 3 clinics can be processed in parallel inside that one process (WORKER_POOL_SIZE=3).

```
cair_service.py (every 5 min)
|
+-- Dev: CLINIC_DB_* -> ClaudMD_Development_Sithum
+-- QA:  Master DB -> ClinicSetup -> each clinic DB
|
Find SubmitStatus = 0 or 3 AND EHRHeaders.IsPublish = 1
|
Build HL7 VXU -> SOAP submitSingleMessage -> CAIR2
|
UPDATE EHRVaccineThirdPartySubmissions + EHRVaccines
```

2.1 How Workers Work (3 Layers)

Layer	File	Role
1 — Scheduler	cair_service.py	Starts once, runs forever. Every SCHEDULER_INTERVAL_SECONDS (300s = 5 min) calls coordinator.run_once(), then sleeps.
2 — Coordinator	worker/coordinator.py (CairCoordinator)	Finds clinics with pending work. Dispatches each clinic to a thread in ThreadPoolExecutor (WORKER_POOL_SIZE=3).

Layer	File	Role
3 — Processor	worker/processor.py (SubmissionProcessor)	Per clinic: loads up to WORKER_BATCH_SIZE (100) pending rows, processes each submission one by one.

One background job, not one per clinic. A single cair_service.py process handles all clinics. Inside it, up to 3 clinic batches run in parallel threads.

2.2 Each Run Cycle (coordinator.run_once)

```

cair_service.py
|
CairCoordinator.run_once()
|
[Dev] CLINIC_DB_* -> check has_eligible_work() -> 1 clinic
[QA]  Master DB -> ClinicSetup (Active=1) -> check each clinic
|
ThreadPoolExecutor (max 3 workers)
/      |      \
Clinic A  Clinic B  Clinic C  (only clinics with pending rows)
batch    batch    batch

```

2.3 Per Clinic Batch (process_clinic_batch)

Step	What happens	DB / code
1	Connect to clinic DB	CairSubmissionRepository
2	SELECT up to 100 rows: SubmitStatus=0 or 3, IsPublish=1	get_eligible_submissions()
3	Set SubmitStatus = 4 (PROCESSING) on selected rows	claim_submissions()
4	For each row — process_submission()	SubmissionProcessor

2.4 Per Submission (process_submission)

Step	Action	Result
1	Load patient + vaccine from DB (JOIN Patients, EHRVaccines, etc.)	VxuPayload
2	Build HL7 VXU message	vxu_builder.py
3	POST SOAP 1.2 submitSingleMessage (username, password, facilityID, hl7Message)	soap_client.py
4	Parse ACK (MSA segment)	ack_parser.py — AA / AE / AR
5a	ACK = AA (accept)	SubmitStatus=1, save HL7+ACK, IsSubmitted=1
5b	HTTP error, ACK = AE, or timeout (temp error)	SubmitStatus=3, AttemptCount+1, retry later
5c	Missing SOAP credentials or ACK = AR / max retries (5)	SubmitStatus=2, save ErrorMessage

2.5 Retry Logic

Attempt	Wait before retry	After 5 failures
1	5 minutes	
2	15 minutes	
3	30 minutes	
4	60 minutes	

Attempt	Wait before retry	After 5 failures
5	120 minutes	Mark FAILED (SubmitStatus = 2)

- Rows with SubmitStatus=3 (RETRY) are picked up again after the wait period on the next run cycle.

2.6 Worker Config (.env)

Setting	Default	Purpose
SCHEDULER_INTERVAL_SECONDS	300	Seconds between each run cycle (5 min)
WORKER_POOL_SIZE	3	Max clinics processed in parallel
WORKER_BATCH_SIZE	100	Max submissions per clinic per cycle
MAX_RETRY_ATTEMPTS	5	Retries before marking FAILED

3. CAIR Endpoints & DB (Sithum Dev Now)

CAIR SOAP Endpoints (Onboarding vs Production)

Onboarding / stage (used now): URL in .env (CAIR_SOAP_URL) on **cdph-interop-stage.cdph.ca.gov** — for SF-013259 testing only; HL7 is sent via **SOAP 1.2 submitSingleMessage**. **Production (later):** URL on **cdph-interop-prod.cdph.ca.gov** — switch after CAIR approves go-live.

Environment	Submission endpoint (CAIR_SOAP_URL)
Onboarding / stage (now)	https://cdph-interop-stage.cdph.ca.gov/services/client_Service.client_ServiceHttpSoap12Endpoint
Production (after approval)	https://cdph-interop-prod.cdph.ca.gov/services/client_Service.client_ServiceHttpSoap12Endpoint

SOAP 1.2 envelope (per WSDL)

Item	Value
WSDL (reference)	https://cdph-interop-stage.cdph.ca.gov/CASTG-WS/IISService?WSDL
SOAP version	1.2 (Soap12Endpoint — SOAP 1.1 returns HTTP 500)
Namespace	urn:cdc:iisb:2011
SOAP action	urn:cdc:iisb:2011:submitSingleMessage
Body fields	username, password, facilityID (SF-013259), hl7Message
.env credentials	CAIR_SOAP_USERNAME / CAIR_SOAP_PASSWORD — set and verified
SOAP facilityID	SF-013259 = vendor/integration org (same as MSH-4)

Two CAIR org codes (important)

Field	What it means	Our value
SOAP facilityID / MSH-4	Vendor / integration org (who sends)	SF-013259 — verified
MSH-22 / RXA-11.4	Clinic site org (where vaccine was given)	SF-012218 — PROVIDER_ORG_ID (accepted by CAIR)

- Tested MSH-22 values: SF-013259 rejected (“cannot be a vendor”); CA0054321 rejected (sample doc only); **SF-012218 accepted** — org blocking error cleared.

What DB We Use

Item	Value
Server	10.103.0.211
Database	ClaudMD_Development_Sithum
Tables read	EHRVaccineThirdPartySubmissions, EHRVaccines, EHRHeaders, CheckInsHeader, Patients, ServiceCodes, Providers, DataGroups
Tables updated	EHRVaccineThirdPartySubmissions, EHRVaccines only
Worker inserts?	No — production worker never INSERTs (read + UPDATE only)
MSH-22 source now	.env PROVIDER_ORG_ID=SF-012218 (not in DB yet). Locations has internal Id only — longer term store CairOrgCode per location.

4. What Fields We Send (HL7 from DB)

Source table	Columns used in HL7
Patients	AccountNumber, FirstName, LastName, DOB, GenderId, HomePhone, CellPhone, Email, Address1, City, State, Zip, RaceId, EthnicityId
DataGroups	Race / ethnicity Description → CDCREC codes (PID-10, PID-22)
CheckInsHeader	Id (check-in), CheckInDate, CheckInTime, ProviderId, LocationId
Providers	NationalProviderIdentifier, FirstName, LastName (ORC-12 / RXA-10)
EHRVaccines	LotNumber, VaccineDate, VaccineTime, Dosage, Manufacturer, Route, BodySite, VaccineExpirationDate
ServiceCodes	NDCNumber (dashed in HL7), Description
EHRHeaders	IsPublish = 1 (filter — only published visits)
.env config	SENDING_FACILITY_ID=SF-013259 (MSH-4); PROVIDER_ORG_ID=SF-012218 (MSH-22 / RXA-11.4)

5. What Fields We Update After Send

EHRVaccineThirdPartySubmissions

Column	Purpose
SubmitStatus	0 pending → 4 processing → 1 success / 2 failed / 3 retry
RequestPayload	HL7 message sent
ResponsePayload	CAIR ACK response
SubmittedDateTime	When CAIR accepted
LastAttemptDateTime	Last send attempt time
AttemptCount	Retry counter
ErrorMessage	Error text on fail/retry
ExternalReferenceId	MSH-10 message control ID

EHRVaccines

- **IsSubmitted = 1** on success only.

6. IsPublish = 1 — What It Means

- **EHRHeaders.IsPublish** = whether the visit/chart is published in ClaudMD (provider finalized the EHR).
- **IsPublish = 1** → visit is ready; CAIR worker may pick up vaccines for that check-in.
- **IsPublish = 0** → visit not published; worker skips it even if a queue row exists.
- **It does NOT mean “sent to CAIR.”** It only means the visit is ready to send.

7. DB Columns — Was CAIR Sent or Not?

Main column: `EHRVaccineThirdPartySubmissions.SubmitStatus`

Value	Meaning
0	Not sent yet (pending)
1	Sent successfully to CAIR
2	Failed permanently
3	Failed temporarily — will retry
4	Currently sending

Also on EHRVaccineThirdPartySubmissions:

- RequestPayload — HL7 sent (empty = not sent yet)
- ResponsePayload — CAIR ACK response
- SubmittedDateTime — when CAIR accepted it
- ErrorMessage — error if failed

Secondary flag: EHRVaccines.IsSubmitted

- 0 or NULL → vaccine not reported to CAIR yet.
- 1 → vaccine successfully sent to CAIR (worker sets on success).

```

EHRHeaders.IsPublish = 1          -> Visit is ready (gate before send)
+
EHRVaccineThirdPartySubmissions.SubmitStatus = 0 -> Queued for CAIR
|
Worker sends HL7
|
SubmitStatus = 1 + EHRVaccines.IsSubmitted = 1 -> Sent to CAIR

```

Quick rule: check **SubmitStatus** on EHRVaccineThirdPartySubmissions — that is the main “CAIR sent or not” column.

8. Status Right Now

Check	Current state
Test data in Sithum DB	Ready (7 submissions seeded)
SOAP envelope	Fixed — SOAP 1.2 (HTTP 500 resolved)
CAIR SOAP credentials	Received and set — login works (HTTP 200)
Vendor org (MSH-4 / SOAP)	SF-013259 — working
Clinic site org (MSH-22 / RXA-11.4)	SF-012218 — accepted (org blocker cleared)
Best test record	Submission 3 — Oak Pereraa (SubmitStatus=1 SUCCESS, MSA AA)
HL7 field fixes	MSH-22/ORC-12.9/ORC-12.21/RXA-10.21 OK; PID-10/11/22 fixed in DB
Tried MSH-22 = SF-013259	Rejected — cannot be a vendor
Tried MSH-22 = CA0054321	Rejected — sample doc code only
Tried MSH-22 = SF-012218	Accepted — no MSH-22 / RXA-11.4 org error
Latest ACK (submission 3)	MSA AA — success; only informational RXA-6 amount note
Remaining ACK issues	None blocking — RXA-6 info only (optional to chase)

Check	Current state
PID-10 / PID-11 / PID-22	Fixed in DB — cleared from ACK
ORC-12.9 assigning authority	Fixed — not in ACK
ORC-12.21 / RXA-10.21 degree	Fixed — Degree empty, Title=MD → sent MD; cleared from ACK
ACK = AA (success)	Yes — submission 3 (2026-09-07)
DB CairOrgCode?	Not on Locations yet — using .env PROVIDER_ORG_ID=SF-012218

9. Next Steps

- Keep .env: SENDING_FACILITY_ID=SF-013259 and PROVIDER_ORG_ID=SF-012218.
- Ask CAIR to review test submissions (especially submission 3 MSA|AA) before production go-live.
- Longer term: EMR adds Locations.CairOrgCode and map CheckInsHeader.LocationId → MSH-22 / RXA-11.4 per site (replace .env for multi-clinic).
- Optional: investigate informational RXA-6 amount warning (0.5 + mL^mL^UCUM still returns Info).

10. Each Field — Source, Required, Status, Sent vs Should Send

Based on current code + latest successful submission **3** (Oak Pereraa, MSA|AA). See Required legend below.
Sent / should send uses latest tested values where applicable.

Required column legend

Code	Meaning
R	Required — must send a value
RE	Required but may be empty — field present; value can be blank
C	Conditional — required only in some cases (e.g. if shot was given / RXA-9=00)
O	Optional — not required by CAIR (we may still send it)

Field mapping

Kind = where the value comes from: **DB** (ClaudMD table.column), **.env** (config), **Fixed** (hard-coded in HL7 builder), or **Code** (default / map in code). Link path for doctor: CheckInsHeader.ProviderId → Providers.

HL7	Kind	Source (table.field or .env)	Req	Status	Sent / should
MSH-3	Fixed	Code — sending application name ClaudMD	O	OK	ClaudMD / keep
MSH-4	.env	.env SENDING_FACILITY_ID (= SF-013259 vendor org)	R	OK	SF-013259 / keep (not for MSH-22)
MSH-6	.env	.env RECEIVING_FACILITY	R	OK	CAIR2 / keep
MSH-9	Fixed	Code — VXU^V04^VXU_V04	R	OK	keep
MSH-11	.env	.env PROCESSING_ID	R	OK	P / keep
MSH-15/16	Fixed	Code — AL AL	R	OK	AL AL / keep
MSH-21	Fixed	Code — Z22^CDCPHINVS	RE	OK	keep
MSH-22	.env	.env PROVIDER_ORG_ID (= SF-012218 clinic site). Later: Locations.CairOrgCode	RE	OK	SF-012218 / keep
PID-3	DB+.env	DB Patients.AccountNumber + .env SENDING_FACILITY_ID (...^^SF-013259^MR)	R	OK	MRN^^^SF-013259^MR / keep
PID-5	DB	DB Patients.LastName, Patients.FirstName (+ Initials if middle)	R	OK	Last^First / keep
PID-7	DB	DB Patients.DateOfBirth	R	OK	YYYYMMDD / keep
PID-8	DB+Code	DB Patients.GenderId → code map (1=M, 2=F, 3=U, 4=X)	R	OK	M/F/U/X / keep
PID-10	DB+Code	DB Patients.RaceId → DataGroups.Description → CDCREC map (cdc_codes)	RE	OK	2106-3^^CDCREC when RaceId set
PID-11	DB	DB Patients.Address1, City, State, ZipCode (skip placeholder address)	RE	OK	street^^city^ST^zip^^H

HL7	Kind	Source (table.field or .env)	Req	Status	Sent / should
PID-13	DB	DB Patients.HomePhone, CellPhone, Email	RE	OK	at least one contact
PID-22	DB+Code	DB Patients.EthnicityId → DataGroups.Description → CDCREC map	RE	OK	2186-5^^CDCREC when set
PD1-12	Code	Code default N (no DB consent column yet)	R	OK	N / keep
PD1-13	DB	DB EHRVaccines.VaccineDate (else CheckInsHeader.CheckInDate)	C	OK	vaccine date / keep
ORC-1	Fixed	Code — RE	R	OK	RE / keep
ORC-3 / ORC-4	DB+.env	DB EHRVaccines.CheckInId (= CheckInsHeader.Id) + .env SENDING_FACILITY_ID	RE	OK	checkinId^SF-013259 / keep
ORC-12	DB	DB via CheckInsHeader.ProviderId → Providers: NationalProviderIdentifier, LastName, FirstName; .21 from Degree (else short Title e.g. MD). .9 fixed NPPES&OID&ISO in code	RE	OK AA	...NPI^^^^^^MD (Degree empty, Title=MD)
RXA-3	DB	DB EHRVaccines.VaccineDate + VaccineTime (else CheckInsHeader.CheckInDate/Time)	R	OK	YYYYMMDD / keep
RXA-5	DB	DB ServiceCodes.NDCNumber (via EHRVaccines.ServiceCodeId); dashed in code	R	OK	58160-0821-11^^NDC / keep
RXA-6	DB	DB EHRVaccines.Dosage	R	OK*	0.5 (CAIR may Info-warn)
RXA-7	Code	Code — mL^mL^UCUM when dose ≠ 999; else blank	C	OK	mL^mL^UCUM / keep
RXA-9	Fixed	Code — 00^NEW IMMUNIZATION RECORD^NIP001	R	OK	keep
RXA-10	DB	Same as ORC-12 — Providers via CheckInsHeader.ProviderId (NPI, name, Degree/Title → .21)	C	OK AA	same ...^^^^^^MD
RXA-11.4	.env	.env PROVIDER_ORG_ID (same as MSH-22)	C/R	OK	^^^SF-012218 / keep
RXA-15	DB	DB EHRVaccines.LotNumber	C/R	OK	lot / keep
RXA-20/21	Fixed	Code — CP A	C/R	OK	CP A / keep
OBX VFC	Code	Code default V01^Not VFC eligible^HL70064 (no DB VFC column yet)	R	OK	V01 / keep
OBX funding	Code	Code default PHC70^Private^CDCPHINVS (2nd OBX)	RE	OK	PHC70 / keep

• **AA achieved** on submission 3. Professional suffix (.21): use **Providers.Degree** if set; if Degree is empty, use **Providers.Title** only when it is a short credential (MD, NP, RN, ...). Example: Degree empty + Title=MD

→ HL7 ends with ^^^^^^^MD.

11. Latest Test — Submission 2 (after PID DB fixes)

Patient **97** (BAGYA ADIKARI) updated in DB: Address1=123 Main St, City/State/Zip=Los Angeles/CA/90210, RaceId=1082 (White), EthnicityId=1084 (Not Hispanic or Latino). Resent submission **2**. MSH-4 = **SF-013259**, MSH-22 / RXA-11.4 = **SF-012218**. SOAP HTTP 200. ACK = **MSA|AE**. PID-10 / PID-11 / PID-22 / ORC-12.9 **cleared** from ACK.

11.1 HL7 VXU message sent

```
MSH|^~\&|ClaudMD|SF-013259||CAIR2|20260908060823+0000||VXU^V04^VXU_V04|59eabeb3-a0b3-4174-a7a8-9a6c2d06c991|1||100064^^^SF-013259^MR||BAGYA ^ADIKARI^^^^^L|20000714|F||2106-3^^CDCREC|123 Main St^^Los Ang
PD1|||||||||N|20260816||||
ORC|RE||1595^SF-013259|1595^SF-013259||||||1093461063^noor^brian^^^^^NPPES&2.16.840.1.113883.4.6&
RXA|0|1|20260816|58160-0842-52^^NDC|0.5|mL^^UCUM||00^NEW IMMUNIZATION RECORD^NIP001|1093461063^noor
OBX|1|CE|64994-7^Vaccine funding program eligibility category^LN|1|V01^Not VFC eligible^HL70064||||
OBX|2|CE|30963-3^Vaccine funding source^LN|1|PHC70^Private^CDCPHINVS||||F||||20260816
```

11.2 CAIR ACK

```
MSA|AE|59eabeb3-a0b3-4174-a7a8-9a6c2d06c991
ERR||RXA^1^10^21|...|W|...|Administering Provider degree missing from RXA-10.21.
ERR||ORC^1^12^21|...|W|...|Ordering Provider degree missing from ORC-12.21.
ERR||RXA^1|...|I|...|Incoming Immunization already exists in the system.
```

11.3 ACK items

Field	Sev.	What we sent	How should send	Source / what to do
ORC-12.9	OK	NPPES&2.16.840.1.113883.4.6 &ISO	Same (keep)	Code — fixed
MSH-22 / RXA-11.4	OK	SF-012218	SF-012218 (keep)	.env PROVIDER_ORG_ID
PID-11.1	OK / Fixed	123 Main St^^Los Angeles^CA^90210^^H	Same (keep)	DB — Patients.Address1/City/State/Zip updated
PID-10	OK / Fixed	2106-3^^CDCREC	Same (keep)	DB — RaceId=1082 (White)
PID-22	OK / Fixed	2186-5^^CDCREC	Same (keep)	DB — EthnicityId=1084 (Not Hispanic or Latino)
RXA-10.21	Warn	1093461063^noor^brian^^^^ NPPES&...&ISO^^^NPI (no .21)	...NPI^^^^^^MD (title = MD/NP/RN/PA/DO)	DB — Providers degree; map to .21
ORC-12.21	Warn	Same provider — no .21	...NPI^^^^^^MD (same title)	DB — same Providers degree; map to .21
RXA	Info	Retry of same immunization	Use a new vaccine/submission for clean AA test	CAIR — dose already stored from earlier send

- **Title** = credential after name (MD, NP, RN, PA, DO), e.g. brian noor, **MD**.
- Next: add provider degree in DB + HL7 map, then send a **new** submission for ACK = **AA**.

12. Latest Test — New Message (Submission 3) provider degree + AA

Sent a **new** submission (Id 3) — not a retry of #2/#6/#7. Patient Oak Pereraa, NDC **58160-0821-11**, provider brian noor (NPI 1093461063). Providers row: **Degree** empty, **Title**=MD — so HL7 ORC-12.21 / RXA-10.21 sent **MD** (prefer Degree; else short Title credential only). Dosage **0.5** with unit **mL^mL^UCUM**. MSH-4 = **SF-013259**, MSH-22 / RXA-11.4 = **SF-012218**. SOAP HTTP 200. ACK = **MSA|AA**. Sithum DB: **SubmitStatus=1 (SUCCESS)**.

12.1 HL7 VXU message sent (RXA / provider focus)

```
MSH|^~\&|ClaudMD|SF-013259||CAIR2|...|VXU^V04^VXU_V04|e53c9776-48dd-431a-82bf-ffb6cc8ee086|P|2.5.1|
PID|1||100075^^^SF-013259^MR||Pereraa^Oak...|...|...|2106-3^^CDCREC|123 Oak Street^^Springfield^IL
PD1|||||||||N|...
ORC|RE||1649^SF-013259|1649^SF-013259|...|1093461063^noor^brian^^^^^NPPES&2.16.840.1.113883.4.6&
RXA|0|1|20260816|58160-0821-11^^NDC|0.5|mL^mL^UCUM||00^NEW IMMUNIZATION RECORD^NIP001|1093461063^no
OBX|1|CE|64994-7^...|1|V01^Not VFC eligible^HL70064|...
OBX|2|CE|30963-3^...|1|PHC70^Private^CDCPHINVS|...
```

12.2 CAIR ACK

```
MSA|AA|e53c9776-48dd-431a-82bf-ffb6cc8ee086
ERR||RXA^1^6|...|I|...|Informational: RXA-6 Administered amount is invalid.
```

12.3 ACK items

Field	Sev.	What we sent	How should send	Source / note
ORC-12.21 / RXA-10.21	OK	...NPI^^^^^^MD	Send one suffix: Degree first, else short Title	From Providers table for the visit doctor. Rule: (1) if Degree filled → use it; (2) else if Title is short credential (MD/NP/RN...) → use Title; (3) else blank. This test: Degree empty + Title=MD → sent MD
RXA-6	Info	0.5	0.5 (keep) or 999 if unknown	Informational only — does not block AA
RXA-7	OK	mL^mL^UCUM	mL^mL^UCUM (CAIR PDF)	Code
RXA-5	OK	58160-0821-11^^NDC	Same (keep)	ServiceCodes.NDCNumber dashed 5-4-2
MSH-22 / RXA-11.4	OK	SF-012218	Same (keep)	.env PROVIDER_ORG_ID
ORC-12.9	OK	NPPES&OID&ISO	Same (keep)	Code — assigning authority
MSA	AA	MSA AA	MSA AA (success)	Sithum SubmitStatus=1 SUCCESS

- **How we take Degree and Title (for ORC-12.21 and RXA-10.21):** Both come from the visit doctor in **Providers**. We send only **one** value after the NPI: (1) use **Degree** if it has a value; (2) if Degree is empty, use **Title** only when Title is a short credential like MD, DO, NP, RN, PA (not a long job name like Physical Therapist); (3) if neither works, leave .21 empty. Example this test — brian noor: Degree empty, Title=MD → HL7 ...NPI^^^^^^MD in both ORC-12 and RXA-10.

- That MD suffix cleared the ORC-12.21 / RXA-10.21 warnings. First clean **MSA|AA** on a new dose. RXA-6 info remains non-blocking.

12.4 Possible Info / Warn / Error after latest code (when data is wrong or empty)

With current code, mapping is correct. CAIR can still return **Info**, **Warn**, or **Error** if clinic DB values are empty, placeholders, or not matching CAIR rules. Code bugs for org / NPI assigning authority / Degree&Title / dashed NDC / race map are already fixed. Check the cause column below.

Field / ACK	Sev.	Seen after latest code?	Cause type	DB / .env / CAIR detail	What to do
RXA-6 amount	Info	Yes — still on AA (sub 3) with 0.5	CAIR side (not empty DB)	DB EHRVaccines.Dosage=0.5 was sent with RXA-7 mL^mL^UCUM. CAIR still returns Info sometimes.	No code change required. Optional: ask CAIR. If dosage unknown, send 999 and blank RXA-7.
ORC-12.21 / RXA-10.21 degree	Warn	Only if DB empty / not usable	DB empty or Title not matching	Providers.Degree empty AND Title empty OR Title is long job name (e.g. Physical Therapist) OR CheckInsHeader.ProviderId null (no Providers join)	Fill Degree or short Title (MD/NP/RN). Set ProviderId on the visit.
PID-10 race	Warn	Only if DB empty / not matching	DB empty or text not mapped	Patients.RaceId null/0, or DataGroups.Description not in our CDCREC map	Set RaceId to a known DataGroups value (e.g. 1082 White → 2106-3).
PID-22 ethnicity	Warn	Only if DB empty / not matching	DB empty or text not mapped	Patients.EthnicityId null/0, or Description not mapped	Set EthnicityId (e.g. 1084 → 2186-5).
PID-11 address	Warn	Only if DB empty / bad value	DB empty or not matching street	Address1/City/State/Zip empty, or placeholder (ADDRESS 1*), or non-street text	Enter a real street address or leave blank.
PID-13 phone/email	Warn / reject risk	Only if DB empty	DB empty	Patients.HomePhone, CellPhone, Email all empty	Fill at least one contact (prefer cell + email).
RXA-5 NDC	Error	Yes if NDC digits wrong (sub 5)	DB value not matching CAIR NDC	ServiceCodes.NDCNumber wrong length/format (e.g. 10-digit 5816082152 → bad dash 5816-0821-52)	Use valid 11-digit NDC in ServiceCodes (CAIR accepted 58160-0821-11).
RXA duplicate dose	Info	Yes if same shot resent	CAIR already has immunization	Not a DB empty issue — retry of same patient/NDC/date/lot	Use a new vaccine/submission for clean AA.
Provider NPI missing	Warn	Only if DB empty	DB empty	Providers.NationalProviderIdentifier blank or ProviderId not set on check-in	Fill NPI on Providers; link ProviderId.

• **On submission 3 (latest AA):** only RXA-6 **Info** appeared. That is **not** because Dosage was empty — DB had 0.5. All Warn/Error rows above are avoided when patient, provider, NDC, and .env are filled correctly.